Homework 3
(due April 13 @15:55, in Blackboard)

1) Derive the dual of the following LPs. Also compute the dual of the dual for each case and interpret your findings.
   a) minimize $c^T x$
      subject to $Ax \succeq b$
   b) minimize $c^T x$
      subject to $Ax \succeq b, x \succeq 0$

2) Problem 5.13 from the reference book.

3) Consider the problem

$$\begin{array}{ll} \text{minimize} & x_1 \\ \text{subject to} & \sqrt{x_1^2 + x_2^2} \leq x_2 \\ & -x_1 \leq 1 \end{array}$$

   a) Calculate the optimal value and the set of optimal solutions.
   b) Derive the dual function and calculate the dual optimal solution. Does strong duality hold? Why? [*Hint:* show that $\lim_{x \to +\infty} (\sqrt{a + x^2} - x) = 0$]

4) Consider the QP (over x_1, x_2):

$$\begin{array}{ll} \text{minimize} & x_1^2 + 2x_2^2 - x_1 x_2 - x_1 \\ \text{subject to} & 5x_1 + 76x_2 \leq 1 \\ & x_1 + 2x_2 \leq u_1 \\ & x_1 - 4x_2 \leq u_2 \end{array}$$

   a) Use CVXPY to find the primal/dual optimal values for $u_1 = -2, u_2 = -3$ and verify the KKT conditions (within reasonable numerical accuracy).
   b) Use these values to predict the optimal value for $u_1 = -2 + \delta_1, u_2 = -3 + \delta_2$, where $\delta_1, \delta_2 \in \{-0.1, 0.1\}$ (four combinations). Verify that $p^\star_{\text{pred}} \leq p^\star_{\text{exact}}$ and assess the prediction gap.

5) Derive the duals for problems 4a, 4b in Hw 2, solve them and verify the KKT conditions.

The following problem is for *extra credit*.

6) Consider the projection of a point on the ℓ_1 unit ball:

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}\|x - a\|_2^2 \\ \text{subject to} & \|x\|_1 \leq 1 \end{array}$$

Derive an efficient method for the dual problem and show how to use it to solve the primal. Compare the run-time of your method vs. a CVXPY solution for randomly chosen $a \in \mathbf{R}^{300}$.
[*Hint:* Show the dual function is differentiable and consider two cases: a) $\|a\|_1 \leq 1$ and b) $\|a\|_1 > 1$]